

Large Language Model Empowered by Domain-Specific Knowledge Base for Industrial Equipment Operation and Maintenance

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Abstract—Industrial equipment operations and maintenance (IEOM) refers to ensuring the normal, safe, and reliable operation of industrial facilities, which covers condition monitoring, equipment maintenance, troubleshooting, repair and maintenance, and system optimization. Currently, the advancements in artificial intelligence have greatly improved the efficiency and effectiveness of IEOM. However, its robustness and generalization in practical applications still need to be improved. Large language models (LLMs) like ChatGPT have recently made breakthrough progress, demonstrating highly intelligent language comprehension capabilities. Therefore, they are expected to drive a new round of transformation in IEOM, promoting the automation and intelligence of the entire IEOM process. However, when using LLMs for practical industrial applications, existing LLMs have fatal limitations as they severely lack domain-specific expertise. This makes it difficult for LLMs to handle technical issues in the industrial field. To this end, this study explores a new solution: LLMs empowered by domain-specific knowledge base (LLM-DSKB). This paper provides a detailed introduction to the core components and implementation details of LLM-DSKB, including the knowledge base, text embedding, vectorized retrieval, etc. The performance of LLM-DSKB is analyzed using real industrial cases, and the results demonstrate that LLM-DSKB can provide more accurate, specific, and industrially relevant results compared to traditional LLMs. This solution will drive the application of LLMs in the industrial field, significantly enhancing the efficiency, effectiveness, and quality of IEOM.

Keywords—Industrial Equipment Operations and Maintenance, Large Language Model, Domain-Specific Knowledge Base

I. INTRODUCTION

Complex industrial equipment plays a crucial role in modern society. With the advancement of technology and the development of industrialization, various complex equipment is widely used in various industries, including transportation, manufacturing, energy, and chemical. These equipment types range from large-scale machinery to automated control systems and others. As the complexity of equipment increases, ensuring their normal operation and high performance through daily

maintenance becomes particularly important [1, 2]. Industrial Equipment operations and maintenance (IEOM) aims to guarantee the normal, safe, and reliable operation of complex industrial equipment, thereby improving industrial operational efficiency, equipment lifespan, and personnel safety [3]. However, considering the complexity and specialization of industrial equipment, many experienced operators and maintenance personnel are required to carry out tasks such as data analysis, real-time condition monitoring, troubleshooting, and maintenance. Additionally, industrial equipment generates massive data during operation, making it impractical to rely solely on manual processing by engineers. Therefore, constructing automated and intelligent IEOM systems is critical for modern industrial enterprises.

In the past 20 years, the development of technologies such as artificial intelligence (AI) has greatly propelled the automation of IEOM [4-6]. This AI-based approach enables automated data analysis and accurate predictions based on learning and induction from historical data [7, 8]. As a result, they have gained extensive research and application in the IEOM field [9-11]. For example, Wang et al. [12] proposed a brain-inspired spiking neural network for the health monitoring of autonomous vehicles. Using a dynamic convolution-based model, Wei et al. [13] achieved accurate semiconductor wafer fabrication monitoring.

Although existing AI-based models have achieved significant results in IEOM, their application scenarios are relatively limited, mainly demonstrating potential in data analysis and predictive tasks. Moreover, the generalization and robustness of current AI-based models are still insufficient to fully meet the demands of practical applications. Therefore, IEOM still requires many experienced engineers to perform tasks such as condition monitoring, troubleshooting, and maintenance to ensure industrial systems' safe and reliable operation. Due to the complexity and specialization of the equipment, these tasks typically require high levels of expertise and extensive experience. This poses high workforce costs for modern enterprises, making it worthwhile to explore methods to reduce the cost of various IEOM processes while ensuring the safety of industrial systems.

Large language models (LLMs) [14], represented by ChatGPT [15], have recently made revolutionary

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breakthroughs in natural language processing (NLP). With impressive language comprehension capabilities and conversational abilities approaching human-like interactions, LLMs have provided a glimpse of the potential for achieving general artificial intelligence. ChatGPT has already sparked innovation in various domains as an outstanding example of LLMs [16]. For instance, ChatGPT can assist doctors in disease diagnosis, medical data analysis, and report generation in the healthcare industry. In the legal field, ChatGPT can be used to provide quick legal consultations or self-service legal assistance. Therefore, LLMs are also expected to be applied in various processes of IEOM, enhancing the automation and intelligence levels of IEOM, reducing enterprise costs, and improving equipment safety [17]. For example, LLMs can be integrated with sensor networks and monitoring systems to monitor device status in real-time and generate corresponding alerts. LLMs can be used to develop intelligent operator assistance systems, providing operators with real-time suggestions and decision support. They can analyze equipment status, operational history, and best practices to offer operational recommendations, optimize parameter settings, and guide the operational process, thereby helping operators improve efficiency and accuracy. By analyzing extensive historical data and device features, LLMs can assist engineers in quickly identifying the causes of equipment failures and providing solutions. It is evident that LLMs have vast application prospects in the field of IEOM, bringing substantial opportunities and benefits to this domain.

However, existing LLMs are trained on massive publicly available text datasets, which poses significant limitations in practical industrial domains. Firstly, LLMs lack domain-specific expertise. Specialized knowledge in the industrial domain, such as equipment technical details, maintenance plans, and guidelines, is often confidential. This makes it challenging for LLMs to grasp the core technical knowledge of the industrial domain, resulting in poor performance when handling professional technical issues. Secondly, the knowledge acquired by LLMs cannot be updated in real-time. For example, ChatGPT's training dataset only includes information up until 2021, so it cannot provide accurate responses to information from the past one or two years. Similarly, in the industrial domain, technical details, maintenance plans, and guidelines for equipment undergo continuous updates and optimizations. If LLMs cannot keep up with this real-time information, it may lead to incorrect decision-making. Fine-tuning LLMs can partially alleviate this issue. However, the confidentiality of industrial data poses challenges to this solution as it may risk data privacy leakage.

To address the aforementioned limitations, this study explores a promising solution by combining LLMs with domain-specific knowledge bases (DSKBs) to empower LLMs with domain expertise. To this end, the study introduces LLM empowered by DSKBs (LLM-DSKB) for IEOM. LLMs possess powerful language comprehension, summarization, and logical reasoning abilities, while DSKBs contain rich and comprehensive domain-specific knowledge. LLM-DSKB maximizes the strengths of LLMs and DSKBs through deep integration. Furthermore, DSKBs can be updated in real-time to ensure that LLM-DSKB stays up-to-date with the latest technological developments. This enables LLM-DSKB to provide professional, specific, and practical results in various industrial tasks. Moreover, DSKBs can be stored offline and locally, eliminating concerns regarding data privacy breaches.

This paper provides a detailed demonstration of the core components and implementation details of LLM-DSKB. Additionally, it validates its considerable potential for application in the industrial domain through real-world case studies.

II. LARGE-SCALE LANGUAGE MODELS

LLMs [14], represented by ChatGPT [15], are a type of language model with a massive number of parameters trained on large-scale datasets. These language models are mostly based on the Transformer architecture and cutting-edge NLP technologies, and they can learn the statistical patterns, semantic relationships, and contextual understanding of language. The core components of LLMs include Transformers and their variants, self-supervised learning, and multimodal fusion. Below is a brief introduction to these core components.

Transformer and Its Variants [18]. The Transformer is a neural network architecture based on attention mechanisms for processing sequential data. It establishes global dependencies in the input sequence through self-attention mechanisms, eliminating the sequential processing limitations of traditional recurrent neural networks. This enables the model to perform parallel computations and better capture long-term dependencies. Assuming the input of the self-attention is $X = [x_1, x_2, \dots, x_k]$, three linear layers with weights $\mathbf{W}^Q, \mathbf{W}^K, \mathbf{W}^V$ are used to encode it into Q (query vector), K (key vector), and V (value vector), respectively. Self-attention can be expressed as:

$$Att(Q, K, V) = Softmax\left(\frac{QK^T}{\sqrt{d}}\right)(V) \quad (1)$$

Furthermore, to encode semantic information in multiple subspaces, the multi-head attention (MHA) mechanism is proposed, which consists of multiple parallel self-attentions, which can be expressed as:

$$O = Linear_h(MHA(X)) = Concat(head^1, head^2, \dots, head^H)\mathbf{W}^H, \quad (2)$$

where $head^H = Att(Q_H, K_H, V_H)$

where $head^H$ represents an attention head (self-attention), H represents the number of attention heads, and $Concat(\cdot)$ represents the splicing of the output results of H attention heads.

Self-Supervised Learning [19]. Self-supervised learning aims to learn valuable representations or features from unlabeled data without manually annotated labels. It achieves this by designing tasks that generate virtual labels, allowing the model to predict them during learning. The key idea behind this approach is to leverage the data's inherent structure and statistical regularities, enabling the model to self-supervise and self-train using the generated labels. Given a set of unlabeled dataset $D = \{X_i\}_{i=0}^N$, the corresponding pseudo label P_i is constructed based on the characteristics of the data itself. This process does not include any manual annotations. Based on the above data, the training loss function is defined as:

$$\mathcal{L}(D) = \min_{\theta} \frac{1}{N} \sum_i^N loss(\mathcal{F}(X_i), P_i) \quad (3)$$

$\mathcal{F}(\cdot)$ represents the network model, and θ represents the learnable parameters. It is evident that self-supervised learning has the advantage of overcoming the challenges and costs of

data annotation, and it can utilize a large amount of unlabeled data for training, thereby improving the effectiveness and adaptability of the model.

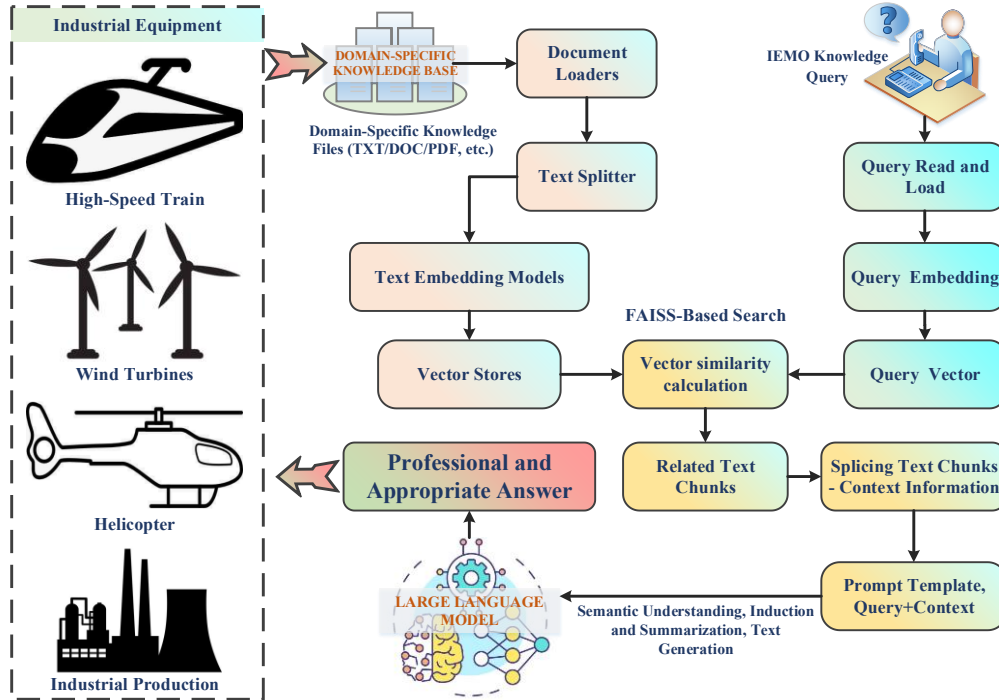


Fig.1 The overall system architecture of the large language model empowered by domain-specific knowledge base.

Multimodal Fusion [20]. Multimodal fusion refers to integrating information from different perceptual modalities (such as images, text, audio, etc.) to obtain a more comprehensive and rich understanding and expressive capability. The key challenge of multimodal fusion lies in effectively combining the heterogeneity and complementarity of different modalities and handling the correlations between modalities. One viable approach is the multimodal attention mechanism, which aims to automatically learn the importance of each modality in different contexts and dynamically adjust the weights between modalities based on their contributions to the task. Suppose the multimodal data is $M = [X_1, X_2, \dots, X_n]$, where X_i represents the i -th modality data, and n represents the number of modalities. Each modality data X_i can be encoded in different ways, for example, $Y_i = \varphi_i(X_i)$, where φ_i represents the feature extraction and encoding function for the i -th modality. Attention-based modality fusion can be expressed as:

$$Z = ATT(\varphi_1(X_1), \varphi_2(X_2), \dots, \varphi_n(X_n)) \quad (4)$$

Multimodal fusion provides a powerful tool for cross-modal information perception of LLMs.

III. LLM EMPOWERED BY DSKB

A. Overall System Architecture

As mentioned above, LLMs, represented by ChatGPT, have the potential to revolutionize the IEMO field with their powerful capabilities, driving the development of automated and intelligent IEMO systems. However, the lack of domain-specific knowledge and the difficulty in updating knowledge in

real-time pose significant challenges for existing LLMs in real-world industrial applications. To this end, this study explores knowledge-base-enabled LLMs to address this limitation.

Fig. 1 illustrates the overall system architecture of LLM-DSKB. Modern industrial systems, such as high-speed trains and wind turbines, inevitably contain massive amounts of technical documents, maintenance records, and manuals, providing the foundation for building secure and offline domain-specific databases. A comprehensive and professional knowledge base can be constructed based on various technical documents and data from industrial systems. After constructing the knowledge base, the LLM-DSKB mainly involves the following processes: document/query reading, text segmentation, text/query embedding, vectorized retrieval, similarity calculation, LLM-based summary, reasoning, and calculation.

Document/query reading utilizes multiple format text readers to handle files in different formats (TXT, DOC, PDF, etc.) and load them into the system. To facilitate the processing of information, the text needs to be split into multiple text chunks that carry semantic meaning. Next, the text embedding model [21] is adopted to encode the text chunks into continuous vectors that computers can understand and process. The text embedding model is a technique to transform text data into low-dimensional continuous vectors, which are used to represent the semantic and grammatical features of the text. Finally, to realize the processing and analysis of large-scale text vectors, it is necessary to construct a vector database and establish corresponding indexes. In this regard, Facebook AI Similarity Search (FAISS) [22] is employed, which is a high-

performance similarity search library developed by the Facebook AI research team. It aims to address the problem of similarity matching in large-scale datasets, particularly excelling in similarity search tasks in vector spaces. With the aid of FAISS, LLM-DSKB can quickly and accurately query and retrieve text segments relevant to the given query, providing precise contextual information to the language model.

On the other hand, LLM-DSKB receives query texts from engineers, which also undergo text reading and vectorization processes. This process encodes the input text into vectors with specific semantic information. Then, using FAISS, it queries and retrieves the most relevant technical information from the vector knowledge base based on the query. Generally, FAISS will return the K professional technical information most relevant to the query, and K can be adjusted. Subsequently, this related professional information and the query are concatenated together based on prompt templates, serving as input to the language model. Finally, leveraging its strong semantic understanding, summarization, induction, and reasoning abilities, the language model provides accurate and appropriate responses based on the given information. This process effectively integrates LLMs with the knowledge base, maximizing their advantages and providing valuable guidance for industrial operations and maintenance.

As shown in **Fig. 1**, in LLM-DSKB, the framework's core components are the text embedding model, vector similarity retrieval, and LLMs. In order to demonstrate the technical solution of LLM-DSKB more clearly, these components are described in detail below.

B. Text Embedding Model

The text embedding model is a technique that encodes text into low-dimensional continuous vector representations by considering the text's context, grammatical patterns, and semantic correlations. It effectively captures the semantic and syntactic features of the text. Text embedding models aim to bring text with similar semantic meanings closer to each other in the embedding space, while dissimilar text remains far apart. This continuous embedding representation enables convenient computation and comparison of text data, thus playing a crucial role in various NLP tasks. Text embedding models can be built based on word vector models or deep learning models such as ELMo [23]. For example, by extracting context-dependent representations from a bidirectional language model, the ELMo [23] model obtains context-specific representations. To encode both the left and right contexts, the model employs a forward L -layer LSTM and a backward L -layer LSTM. These LSTM layers are used to create contextualized representations at each layer j , which are obtained by concatenating the left-to-right and right-to-left representations. As a result, the ELMo model generates N hidden representations ($h_{1,j}, h_{2,j}, \dots, h_{N,j}$) for a sequence of length N . To use ELMo in downstream tasks, the $(L+1)$ -layer representations for each token k are aggregated as:

$$ELMo_k^{task} = \gamma^{task} \sum_{j=0}^L s_j^{task} h_{k,j} \quad (5)$$

The layer-wise weights, denoted as s^{task} , are normalized by the softmax function to linearly combine the representations of the

token k from the $(L+1)$ -layer. Additionally, γ^{task} represents a task-specific constant.

C. Vector Similarity Retrieval

Vector similarity retrieval is the core of LLM-DSKB, as it determines whether LLMs can obtain accurate and comprehensive domain-specific expertise. After encoding the knowledge base and query as vectors, the goal of vector-based similarity retrieval is to find the most relevant text information to the query. In NLP, cosine similarity is commonly used to measure the similarity between two vectors in semantic space. Assuming b represents the vector representation of text in the knowledge base and q represents the vector representation of the query, their cosine similarity can be expressed as follows:

$$\text{cosine}(v, w) = \frac{b \cdot q}{|b| |q|} = \frac{\sum_{i=1}^L b_i q_i}{\sqrt{\sum_{i=1}^L b_i^2} \sqrt{\sum_{i=1}^L q_i^2}} \quad (6)$$

where L represents the length of the vectors. Considering that the knowledge base data can be extremely large, performing retrieval in a simple iterative manner can be time-consuming. Therefore, in LLM-DSKB, FAISS [22] can be introduced to accelerate the speed of similarity search. FAISS provides a set of efficient similarity search algorithms and index structures that can be used to quickly find data information in large-scale vector datasets that is most similar to the query vector. Its main advantage lies in its outstanding performance in handling millions or even billions of vector data in high-dimensional vector spaces.

D. Lightweight Language Model

In the industrial domain, considering the limited computational resources and the requirement for quick response, lightweight language models can better meet the needs of relevant applications. In this study, we employ ChatGLM-6B [24, 25], a lightweight language model developed by Tsinghua University. It is a bilingual language model with 6.2 billion parameters and can be locally deployed on consumer-grade GPUs. In ChatGLM-6B, autoregressive blank infilling serves as its core self-supervised training method, enabling training and learning on large-scale Chinese and English text datasets. Additionally, it incorporates 2D positional encodings and allows arbitrary order to predict spans, achieving more efficient self-supervised pretraining and enhancing the model's language understanding and semantic encoding capabilities. Briefly, given a text sequence $X = [x_1, \dots, x_n]$, we sample text spans $\{S_1, \dots, S_m\}$ from it. Each span S_i represents a consecutive token sequence $[s_{i,1}, s_{i,2}, \dots, s_{i,l_i}]$. ChatGLM-6B replace (or "corrupt") each span with a single mask token to create X_{corrupt} . The objective of the model is to recover the original spans through an autoregressive process. To facilitate interactions among the corrupted spans, their visibility to each other is determined by randomly permuting their order. The pre-training objective [25] is then defined as follows:

$$\mathcal{L} = \max_{\theta} \mathbb{E}_{z \sim Z_m} \left[\sum_{i=1}^m \log \prod_{j=1}^{l_i} p(s_{i,j} | X_{\text{corrupt}}, S_{z < i}, S_{i < j}) \right] \quad (7)$$

Here, Z_m represents the set of all permutations for the

sequence. Additionally, $S_{z < i}$ denotes the sequence $[S_{z_1}, \dots, S_{z_{i-1}}]$.

E. Implementation Details

This study implemented LLM-DSKB on a workstation with an Intel 10900K CPU, GeForce RTX 3090 GPU, and 32GB of memory. LLM-DSKB is, in fact, the result of advancements in cutting-edge language analysis technology and requires support from various techniques. Currently, the LangChain framework¹ provides robust technical support for implementing LLM-DSKB. The LangChain framework offers a range of functionalities, including multi-format text reading, text splitting, text embedding, and vector similarity retrieval. Furthermore, thanks to the development of large-scale pre-trained models, text embedding models such as ELMo [23], and M3E² can provide reliable text embedding techniques for LLM-DSKB. Considering efficiency and resource constraints in local implementation, the M3E model was adopted in this study. Additionally, the FAISS framework provides powerful tools for fast vector retrieval and querying, enabling billion-scale similarity searches on GPUs. Currently, based on the aforementioned technical foundations, projects such as LangChain-ChatGLM³ and Wenda⁴ have implemented question-answering systems based on scalable knowledge

bases and have achieved excellent performance. Based on the existing work, this study further explores the feasibility and performance of DSKB-empowered LLM in industrial operations and maintenance, aiming to drive the research progress of large-scale models in the industrial domain.

IV. INDUSTRIAL CASE STUDIES

In this section, wind turbines were used as an example to validate the performance of LLM-DSKB in a specific industrial domain. For this purpose, a small-scale knowledge base related to wind turbines was constructed, consisting primarily of professional knowledge documents such as "Wind Turbine Troubleshooting Manual," "Wind Turbine Fault Handling Manual," and "Wind Turbine Operation and Maintenance Manual." These professional knowledge resources mainly cover the troubleshooting, operation, and maintenance of wind turbines. Due to the specialized nature of the content, most LLMs lack knowledge in this area. Due to the length constraints of the paper, this study shows the question and answer records of three professional technical questions and compares the difference between LLM-DSKB and LLM reply results to verify the effectiveness of LLM-DSKB. The detailed question-and-answer records are shown in **TABLE I**.

TABLE I
LLM-DSKB AND LLM Q&A RECORDS ON WIND TURBINE RELATED QUESTIONS

Question-1	How to inspect and maintain the filter of the gearbox oil cooling system of a wind turbine?
LLM-DSKB	Maintenance of the filter in the gearbox oil cooling system: (1) Ensure that the unit is in a stopped state and the lubrication and cooling systems are fully depressurized. (2) Close the ball valve between the gearbox and the oil pump. (3) Clean the area around the filter with a cloth and remove the connecting hose and end cap between the filter and the gearbox. (4) Check if the filter element is clogged. If it is clogged, it needs to be replaced. • Replacement process: Open the drain valve at the bottom of the filter, drain the dirty oil, and remove the filter element. Remove the debris collector from the filter and clean it. Reinstall the debris collector, insert a new filter element, close the drain valve, tighten the end cap (after tightening the end cap, loosen it a quarter turn for ease of future operations), connect the vent hose, and open the ball valve. Following the above steps, regular inspection and maintenance of the gearbox oil cooling system filter can be carried out to ensure system stability and reliability.
LLM	The filter of gearbox oil cooling system is an important maintenance component. The following are the steps to inspect and maintain the filter in the gearbox oil cooling system: (1) Inspect the filter: Check if the filter is damaged. A filter testing tool can be used to test the filtering capability. If the filter fails to filter enough oil, it needs to be replaced. (2) Clean the filter: If the filter is damaged and needs to be replaced, cleaning it can be an alternative. Professional filter cleaning tools can be used to clean the filter. When cleaning the filter, pay special attention to the holes on the filter, as these holes are used for oil filtration. If these holes are clogged, it can prevent oil from passing through the filter, affecting the normal operation of the gearbox. (3) Replace the filter: If the filter needs to be replaced, follow the instructions in the manual for the replacement procedure. When replacing the filter, choosing a filter with the same model as the existing one is important to ensure proper functionality after replacement. (4) Inspect the gearbox oil: After replacing the filter, check the quality of the gearbox oil. A filter testing tool can be used to test the filtering capability of the gearbox oil. If the filtering capability of the gearbox oil is poor, it needs to be replaced. (5) Perform regular maintenance: In addition to regularly replacing the filter and checking the quality of the gearbox oil, regular maintenance of the gearbox oil cooling system is necessary. This includes checking if the filter is clogged and replacing the filter and gearbox oil as needed.
Question-2	What is the phenomenon of wind turbine blades after lightning strikes?
LLM-DSKB	The blades after lightning strikes have the following phenomena: (1) Small areas of damage may occur near the blade tip; (2) There are black marks on the surface of the blade, which look like grease or oil stains from a distance; (3) Splitting of the blade tip or edge; (4) There are longitudinal cracks on the surface of the blade; (5) split in the middle of the shell; (6) The blades make a clicking sound as they rotate slowly. These phenomena all indicate that lightning strikes may have damaged the blade. It should be noted that different lightning strike methods and geographical locations may affect blades differently.
LLM	The wind turbine blades after a lightning strike may have the following phenomena: (1) Damage: Lightning strikes can cause damage to blades, causing them to break or deform. (2) Fusing: Some blades will fuse after lightning strikes; that is, after the temperature of some blades rises to a certain level, the melt will fall off from the blades, forming a fusing. (3) Burn damage: Lightning strikes may also cause burn damage to the blade, resulting in damage or cracks on the blade surface. (4) Mechanical failures: Lightning strikes can cause mechanical failures, including damaged bearings, gearbox failures, etc. (5) Safety issues: If the blades are damaged due to a lightning strike, the wind turbine may not work properly and even cause safety hazards, because the damaged blades may fall off or get stuck in the equipment, causing personal injury or other losses. Therefore, in lightning weather, wind turbines need to take corresponding safety measures, including installing lightning arresters, regular maintenance, inspection, etc., to protect the safety of equipment and personnel.
Question-3	Fault code 070022 and its cause.
LLM-DSKB	The reason for the fault code 070022 is that the temperature of the temperature sensor A of the generator bearing exceeds the limit. It is necessary to stop the machine and check the sensor to ensure its normal operation. The fault code 070031 indicates that the generator cooling liquid temperature has exceeded the set limit. It requires a shutdown to inspect the temperature of the cooling liquid and ensure it meets the set value.
LLM	Fault code: 070022 means that the power supply of an electronic component (such as sensor, controller, filter, etc.) is insufficient or the circuit is faulty. The specific cause of the failure may vary depending on the device model, use environment, operation method, and other factors.

As shown in **TABLE I**, we consulted both LLM-DSKB and LLM on technical questions related to "Inspection and Maintenance of Wind Turbine Filters," "Wind Turbine Blade Lightning Strikes," and "Wind Turbine Fault Codes." Comparing the responses of LLM-DSKB and LLM, it can be

observed that LLM's responses are general and lack specific technical details. In contrast, LLM-DSKB's responses are more professional, providing clear and concise technical details that offer practical guidance. Taking Question 1 as an example, the reply content of LLM-DSKB has detailed operation steps and

¹https://python.langchain.com/docs/get_started/introduction.html

³<https://github.com/imClumsyPanda/langchain-ChatGLM>

²<https://huggingface.co/moka-ai/m3e-base>

⁴<https://github.com/wenda-LLM/wenda>

technical details, so it can provide valuable information for industrial practice. However, the content of LLM's reply is broad, redundant, and does not have feasible operational details, because it cannot provide enough valuable information for industrial practice. Additionally, for Question 2, LLM-DSKB's response is more refined and accurate, comprehensively describing the potential damages caused by lightning strikes on wind turbine blades. In contrast, LLM's response appears comprehensive but consists of generic information that is difficult to apply to industrial practice. Furthermore, Question 3 validates the troubleshooting capability of LLM-DSKB. Given a specific fault code, LLM-DSKB can quickly retrieve relevant information, such as the fault category and fault condition, from the knowledge base and provide accurate fault information. LLM, lacking domain-specific knowledge, provides a result unrelated to wind turbines.

In general, LLM-DSKB has improved the problem of lack of professional knowledge in specific fields of ordinary LLMs, and it can give detailed, specific, and industrial-needed responses when facing professional knowledge in the industrial field. This enables LLM-DSKB to be applied in various professional technical fields, leveraging the advantages of LLM and enhancing the effectiveness of IEOM-related applications.

V. CONCLUSIONS

IEOM is dedicated to ensuring the normal, safe, and reliable operation of industrial facilities, covering aspects such as condition monitoring, equipment maintenance, troubleshooting, repair and maintenance, and system optimization. Recently, LLM has made significant breakthroughs in language understanding, text generation, and summarization capabilities, providing hope for changing the application mode of IEOM and promoting automation and intelligent construction. However, due to the lack of industrial expertise in LLM, its practical industrial application is limited. Therefore, this study aims to address practical industrial issues in IEOM by exploring the DSKB-based LLM framework. Additionally, this paper provides a detailed introduction to the core components and implementation details of LLM-DSKB, including the knowledge base, text embedding, and vectorized retrieval. The analysis of real industrial cases shows that LLM-DSKB can provide more accurate, specific, and industry-relevant results compared to traditional LLM. This approach will drive the application of LLM in the industrial field, effectively improving the efficiency, effectiveness, and quality of IEOM.

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